DATA STRUCTURES HOME WORK 1

1)SUM OF TWO MATRICES

CODE:

```
#include<stdio.h>
int main()
{
    int m,n;
    printf("Enter the no of rows:");
    scanf("%d",&m);
    printf("Enter the no of columns:");
    scanf("%d",&n);
    int arr1[m][n],arr2[m][n],sum[m][n];
    printf("Enter the first matrix:\n");
    for(int i=0;i<m;i++)
    {
        for(int j=0;j<n;j++)
        {
            scanf("%d",&arr1[i][j]);
        }
        printf("\n");
    }
    printf("Enter the second matrix:\n");
    for(int i=0;i<m;i++)
    {
        for(int j=0;j<n;j++)
        {
            scanf("%d",&arr2[i][j]);
        }
        printf("\n");
    }
    printf("Sum of the matrices:\n");
    for(int i=0;i<m;i++)
    {
        for(int j=0;j<n;j++)
        {
            sum[i][j]=arr1[i][j]+arr2[i][j];
        }
    }
    for(int i=0;i<m;i++)
    {
        for(int j=0;j<n;j++)
        {
```

```
            printf("%d  ",sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

Enter the no of rows:3

Enter the no of columns:3

Enter the first matrix:

7 7 7 7 7 7 7 7 7

Enter the second matrix:

7 7 7 7 7 7 7 7 7

Sum of the matrices:

14 14 14

14 14 14

14 14 14

---

2)DISPLAYING ARRAY IN REVERSE ORDER

CODE:

```
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the size of the array:\n");
    scanf("%d",&n);
    int arr[n],rev[n];
    printf("Enter the elements of the array:\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    printf("Array in reverse order:\n");
    for(int i=n-1;i>=0;i--)
    {
```

```
        printf("%d\n",arr[i]);
    }
}
```

OUTPUT:

Enter the size of the array:

5

Enter the elements of the array:

7

18

45

31

96

Array in reverse order:

96

31

45

18

7

---

3)COUNTING DUPLICATE ELEMENTS IN AN ARRAY

CODE:

```
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the size of the array:\n");
    scanf("%d",&n);
    int arr[n],dup=0;
    printf("Enter the elements of the array:\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    for(int i=0;i<n;i++)
```

```
    {
        for(int j=i+1;j<n;j++)
        {
            if(i!=j)
            {
                if(arr[i]==arr[j])
                {
                    dup++;
                }
            }
        }
    }
    printf("No of duplicate elements:%d",dup);
    return 0;
}
```

OUTPUT:

Enter the size of the array:

5

Enter the elements of the array:

3

7

7

31

33

No of duplicate elements:1

---

4)PRINT UNIQUE ELEMENT IN AN ARRAY

CODE:

```
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the size of the array:\n");
    scanf("%d",&n);
```

```c
    int arr[n];
    printf("Enter the elements of the array:\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    printf("Unique elements:");
    for(int i=0;i<n;i++)
    {
        int dup=0;
        for(int j=0;j<n;j++)
        {
            if(i!=j)
            {
                if(arr[i]==arr[j])
                {
                    dup++;
                }
            }
        }
        if(dup==0)
            printf("%d\n",arr[i]);
    }
    return 0;
}
```

OUTPUT:

Enter the size of the array:

5

Enter the elements of the array:

7

31

31

77

77

Unique elements:7

---

5)INSERT IN SORTED ARRAY

CODE:

```
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the size of the array:\n");
    scanf("%d",&n);
    int arr[n];
    printf("Enter the elements of the sorted array:\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
    int num;
    printf("Enter the element to be inserted:");
    scanf("%d",&num);
    int j=0;
    for(int i=0;i<n;i++)
    {
        if(num>arr[i])
            j++;
        else
        {
            n++;
            for(int i=n-1;i>j;i--)
            {
                arr[i]=arr[i-1];
            }
            arr[j]=num;
            break;
        }
    }
    printf("Array after inserting:\n");
    for(int i=0;i<n;i++)
    {
        printf("%d\n",arr[i]);
    }
    return 0;
}
```

OUTPUT:

Enter the size of the array:

5

Enter the elements of the sorted array:

7

18

31

33

96

Enter the element to be inserted:45

Array after inserting:

7

18

31

33

45

96

---

6)INSERT IN UNSORTED ARRAY

CODE:

```
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the size of the array:\n");
    scanf("%d",&n);
    int arr[n];
    printf("Enter the elements of the unsorted array:\n");
    for(int i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }
```

```c
    int num,pos;
    printf("Enter the element to be inserted:");
    scanf("%d",&num);
    printf("Enter the position to be inserted:");
    scanf("%d",&pos);
    n++;
    for(int i=n-1;i>=pos;i--)
    {
        arr[i]=arr[i-1];
    }
    arr[pos-1]=num;
    printf("Array after inserting:\n");
    for(int i=0;i<n;i++)
    {
        printf("%d\n",arr[i]);
    }
    return 0;
}
```

OUTPUT:

Enter the size of the array:

5

Enter the elements of the unsorted array:

7

45

18

33

31

Enter the element to be inserted:81

Enter the position to be inserted:4

Array after inserting:

7

45

18

81

33

7)MATRIX MULLTIPLICATION

CODE:

```
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the no of rows and columns:");
    scanf("%d",&n);
    int arr1[n][n],arr2[n][n],mul[n][n];
    printf("Enter the first matrix:\n");
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            scanf("%d",&arr1[i][j]);
        }
        printf("\n");
    }
    printf("Enter the second matrix:\n");
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            scanf("%d",&arr2[i][j]);
        }
        printf("\n");
    }
    printf("Multiplication of the matrices:\n");
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            mul[i][j]=0;
            for(int k=0;k<n;k++)
            {
                mul[i][j]+=arr1[i][k]*arr2[k][j];
            }
        }
```

```
    }
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            printf("%d\t",mul[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

Enter the no of rows and columns:3

Enter the first matrix:

18 18 18 18 18 18 18 18 18

Enter the second matrix:

7 7 7 7 7 7 7 7 7

Multiplication of the matrices:

378 378 378

378 378 378

378 378 378

---

8)MATRIX TRANSPOSE

CODE:

```
#include<stdio.h>
int main()
{
    int m,n;
    printf("Enter the no of rows:");
    scanf("%d",&m);
    printf("Enter the no of columns:");
    scanf("%d",&n);
    int mat[m][n],tp[m][n];
    printf("Enter the matrix:\n");
    for(int i=0;i<m;i++)
```

```
    {
        for(int j=0;j<n;j++)
        {
            scanf("%d",&mat[i][j]);
        }
        printf("\n");
    }
    for(int i=0;i<m;i++)
    {
        for(int j=0;j<n;j++)
        {
            if(i==j)
                tp[i][j]=mat[i][j];
            else
                tp[j][i]=mat[i][j];
        }
    }
    printf("Transpose Matrix:\n");
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<m;j++)
        {
            printf("%d\t",tp[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

OUTPUT:

Enter the no of rows:2

Enter the no of columns:2

Enter the matrix:

7 18

31 11

Transpose Matrix:

7 31

18 11

9)RIGHT DIAGONAL SUM

CODE:

```
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the no of rows and columns:");
    scanf("%d",&n);
    int arr[n][n],sum=0;
    printf("Enter the matrix:\n");
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            scanf("%d",&arr[i][j]);
        }
        printf("\n");
    }
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            if(i+j==n-1)
                sum+=arr[i][n-i-1];
        }
    }
    printf("Sum of the right diagonal elements is %d",sum);
    return 0;
}
```

OUTPUT:

Enter the no of rows and columns:2

Enter the matrix:

92 55

47 53

Sum of the right diagonal elements is 102

---

10)CHECK FOR IDENTITY MATRIX

CODE:

```
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the no of rows and columns:");
    scanf("%d",&n);
    int arr[n][n],flag=0;
    printf("Enter the matrix:\n");
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            scanf("%d",&arr[i][j]);
        }
        printf("\n");
    }
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            if(i==j)
            {
                if(arr[i][j]!=1)
                    flag=1;
            }
            else
            {
                if(arr[i][j]!=0)
                    flag=1;
            }
        }
    }
    if(flag==0)
        printf("Identity Matrix");
    else
        printf("Not an Identity Matrix");
    return 0;
}
```

OUTPUT:

Enter the no of rows and columns:2

Enter the matrix:

1 0

0 1

Identity Matrix

---

11)TO FIND NO OF VOWELS,CONSONANTS,DIGITS AND WHITE SPACES

CODE:

```
#include<stdio.h>
int main()
{
    int vow=0,con=0,dig=0,sp=0;
    char str[100];
    printf("Enter a String:");
    fgets(str,sizeof(str),stdin);
    int i=0;
    while(str[i]!='\0')
    {

if(str[i]=='a'||str[i]=='A'||str[i]=='e'||str[i]=='E'||str[i]=='i'||
str[i]=='I'||str[i]=='o'||str[i]=='O'||str[i]=='u'||str[i]=='U')
            vow+=1;
        else if(str[i]>='0'&&str[i]<='9')
            dig+=1;
        else if(str[i]==' ')
            sp+=1;
        else if(str[i]>='A'&&str[i]<='Z'||str[i]>='a'&&str[i]<='z')
            con+=1;
        i++;
    }
    printf("No of vowels:%d\nNo of consonants:%d\nNo of
spaces:%d\nNo of digits:%d",vow,con,sp,dig);
    return 0;
}
```

OUTPUT:
Enter a String:Yoga Lakshmi 13

No of vowels:4

No of consonants:7

No of spaces:2

No of digits:2

---

12)FREQUENCY OF A CHARACTER IN A STRING

CODE:

```
#include<stdio.h>
int main()
{
    char str[100],ch;
    int count=0;
    printf("Enter a String:");
    fgets(str,sizeof(str),stdin);
    printf("Enter the character:");
    scanf("%c",&ch);
    int i=0;
    while(str[i]!='\0')
    {
        if(str[i]==ch)
            count++;
        i++;
    }
    printf("Frequency of the character:%d",count);
    return 0;
}
```

OUTPUT:

Enter a String:Visalini

Enter the character:i

Frequency of the character:3

---

13)COUNT WORDS IN A STRING

CODE:

```
#include<stdio.h>
int main()
```

```
{
    char str[100];
    int count=0;
    printf("Enter a String:");
    fgets(str,sizeof(str),stdin);
    int i=0;
    while(str[i]!='\0')
    {
        if(str[i]==' ')
            count++;
        i++;
    }
    printf("No of Words in a String:%d",count+1);
    return 0;
}
```

OUTPUT:
Enter a String:Pirates of the Carribean

No of Words in a String:4

---

14)SORT STRING ARRAY

CODE:

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[5][20], temp[20];

    printf("Enter 5 strings:\n");
    for(int i = 0; i < 5; i++)
        scanf("%s", str[i]);

    for(int i = 0; i < 4; i++)
    {
        for(int j = i + 1; j < 5; j++)
        {
            if(strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
```

```
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("Sorted strings:\n");
    for(int i = 0; i < 5; i++)
        printf("%s\n", str[i]);

    return 0;
}
```

OUTPUT:

Enter 5 strings:

Visalini

Yoga

Anu

Achu

Lakshmi

Sorted strings:

Achu

Anu

Lakshmi

Visalini

Yoga

---

15)TOGGLE CASE OF A SENTENCE

CODE:

```
#include<stdio.h>
int main()
{
    char str[100];
    printf("Enter a string:");
    scanf("%s",str);
    int i=0;
```

```
    while(str[i]!='\0')
    {
        if(str[i]>='A'&&str[i]<='Z')
            str[i]+=32;
        else if(str[i]>='a'&&str[i]<='z')
            str[i]-=32;
        i++;
    }
    printf("Toggled String:%s",str);
    return 0;
}
```

OUTPUT:

Enter a string:CrIcKeT

Toggled String:cRiCkEt
